

gateway alpha1: FR relative improvement G(N) at equal budget

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method).
Paired per seed (same N and seed: shared initial conditions and Langevin noise).

